MINI PROJECT – ABSTRACT

Course: Java Programming

Project Type: Java Desktop Application using Swing UI & JDBC

Milestone 1: Project Abstract with Team Details

Marks: 2

1. PROJECT TITLE

Title of the Project: Exam Seat Arrangement System

2. TEAM DETAILS

Sl. No.	Name of Student	Roll No.	Role	HackerRank Profile / Username	Java Stars in HackerRank
1	Antony Zavior K	62	Team Lead	https://www.hackerrank.com/profile/haxey61790	⭐⭐⭐⭐⭐
2	Lijo P Thomas	67	Team Member	https://www.hackerrank.com/profile/lijot707	⭐⭐⭐⭐⭐
3	Milind Krishnan	69	Team Member		

Note: Every team member shall create a HackerRank profile and complete Java practice problems. The number of Java Stars obtained on HackerRank as on the date of submission shall be entered above.

3. PROJECT ABSTRACT

Problem / Motivation

Before every internal or university examination, colleges have to decide which student sits in which hall, row and bench. This is usually done by hand on paper or in spreadsheets. It takes a lot of staff time and is easily affected by mistakes such as students being seated twice, halls being overfilled, or students of the same branch and subject sitting next to each other, which makes malpractice easier. Last-minute changes, such as a hall becoming unavailable or a student being added late, force the whole plan to be redone. A dedicated application is needed to produce a fair, clash-free seating plan quickly and keep it easy to update and share.

Proposed Solution

We propose a Java desktop application built with Swing and JDBC that automates exam seat allocation. The exam cell administrator enters the halls with their row and column layout, the exams with their dates and sessions, and the list of students. The application then allocates seats automatically using rules such as filling each hall up to its capacity and avoiding adjacent seats for students of the same branch or subject. It stores the result in a database and shows it as a visual seating grid. Intended users are the exam coordinator or administrator, invigilators, and students who need to look up their seat.

Major Features

- **Secure admin login:** role-based access for administrator and staff, with password protection.
- **Hall management:** add, edit and remove halls with number of rows, columns and seats per bench.
- **Student and exam management:** maintain student records (branch, semester, register number) and exam schedules, including CSV import of student lists.
- **Automatic seat allocation:** allocates students across halls session-wise while avoiding same-branch or same-subject neighbours and preventing double booking of a seat.
- **Visual seating chart:** a Swing grid for each hall that shows which seat belongs to which student, with manual swap or adjustment.
- **Seat search:** find a student's hall and seat by register number or name.
- **Invigilator assignment:** assign staff to halls for each session.
- **Reports and printing:** hall-wise seating lists, attendance sheets and notice-board seating plans that can be printed or exported.
- **Validation and error handling:** input checks, capacity warnings and clear error messages using exception handling.

Technologies to be Used

- Programming Language: Java
- User Interface: Java Swing
- Database Connectivity: JDBC
- Database: MySQL
- Development Environment / IDE: IntelliJ IDEA

Expected Outcome

A fully working, well-documented Java desktop application that replaces manual seating plans with a fast, accurate and database-backed system, so an exam seating arrangement can be produced in minutes and changed easily. The project will demonstrate object-oriented design (classes, inheritance, interfaces), a layered structure with a DAO pattern over JDBC, Swing GUI development, CRUD operations, exception handling, coding standards and testing, along with a complete project report.

Team Lead Signature: _____ **Date: _____**